COMMERCIAL ITEM DESCRIPTION

CALIBRATOR, VOLTAGE

The General Services Administration has authorized the use of this Commercial Item Description.

SCOPE This Commercial Item Description (CID), describes a precision calibrator, voltage for precise measurement of and calibration of AC voltage and current The calibrator voltage is part of the F-15 AN/TSM-131A Electrical Standards Set (ESS) and is rack-mounted in a transportable operating case. This CID represents a minimum requirement for the F-15 ESS Thermal Transfer Standard in which only those manufacturers who meet or surpass the requirements are qualified. Equipment may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5, style F and color R in accordance with MIL-T-28800.

APPLICABLE DOCUMENTS.

SPECIFICATIONS. The following specification forms a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation This CID makes provisions for bid sample testing in accordance with MIL-T-28800, paragraph 4.2.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

(Copies of specification required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

ORDER OF PRECEDENCE. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and applicable documents.

1. SAFETY. The equipment shall be UL (Underwriters Laboratories) listed and approved and shall comply with the type III safety requirements of MIL-T-28800

2. PERFORMANCE CHARACTERISTICS. The following electrical specifications are minimum requirements.

FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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a. VOLTAGE RANGE. 0.5 to 1,000 V

b. UNCERTAINTY

FREQUENCY (Hz)	VOLTAGE RANGE			
	0.5-10V	20V-50V	100V-500V	1000V
5-20K	±0.01%	±0.01%	±0.01%	±0.02%
20-50K	±0.01%	±0.01%	±0.01%	±0.04%
50K-100K	±0.05%	±0.05%	±0.20%	
100K-500K	±0.10%	±0.10%		
500-1M	±0.10%			

c. SEARCH FUNCTION. Meter display of input as percentage of range.

d INPUT IMPEDANCE. 180 ohms/V

e. POLARITY. Reversible, front-panel push-button.

f. GALVONOMETER RESOLUTION 12 ppm of input/scale division at full range, 60 ppm of input/scale division at half range

g THERMAL COUPLE REVERSAL ERROR. Less than or equal to 100 ppm of input at full range, less than or equal to 300 ppm at half range

h OVERLOAD PROTECTION Withstands up to 1500 V dc or rms AC at any range.

i. POWER 115 V or 230 V AC ±10%, 50 to 60 Hz, 7W self-contained, rechargeable batteries, 200 hours operation, 16 hours recharge time

j SIZE. Not to exceed 7 inches H x 17 inches W x 24 inches D.

k WEIGHT. Not to exceed 50 lbs.

3. COMPATIBILITY. This equipment shall be mechanically, electrically and physically compatible with other equipment and accessories of the ESS as the ESS is designed as a measurement entity. The calibrator voltage shall be capable of rack mounting in a standard 19 inch rack.

4. MANUALS. The manufacturer shall furnish one copy of instruction manuals for each instrument in this equipment group. It shall contain at least the following. Operating Instructions, Theory of Operation, Circuit Description, Schematics, Trouble Shooting Trees, Pictorial Layouts, Parts Locator Diagrams, List of Replacement Parts, Part Numbers, and Manufacturers. In addition, all internal major components, assemblies, subassemblies, and plug-ins, shall be permanently labeled in accordance with manuals, diagrams and schematics furnished.

5. CALIBRATION INTERVAL. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

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CERTIFICATION. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract).

PACKAGING, PACKING AND MARKING. The item shall be packaged in accordance with normal commercial practices and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers shall be marked as specified in the contract or order.

KNOWN ACCEPTABLE PRODUCTS. The following product(s) are known to meet the requirements of this CID.

MANUFACTURER	MODEL	NSN
John Fluke Mfg Co., Inc	540b	6625-00-288-0184

Custodian:
Air Force - 99
Navy - EC
DAL - ES

Preparing activity
Air Force - 82

Review activity:
Army - CR

Project 6625-F850

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